Revisiting Stale TESS Single-transit Community Candidates: Mostly Eclipsing Binaries, Two New Duotransits

Jon McBee (ORCID 0009-0000-7524-4470) · Independent researcher, USA · jon.mcbee@gmail.com

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Abstract

Community TESS Objects of Interest (CTOIs) flagged as single transits are rarely revisited. We re-examined 306 such stars against all TESS data through Sector 108. We used a validated repeat search, our own photometry for sectors not yet processed by the TESS pipelines, difference-image localization and Gaia DR3 binarity diagnostics. Of 272 stars whose catalogued event we recover, 79 show repeat events. Of these, 21 are Gaia-flagged binaries (three Gaia orbits reproduce our periods), 18 are blends, 23 fit no single period, and 17 needed vetting by hand. CTOI 147035057.01 has a fixed 37.315 d period, but a grazing eclipse on an RV-variable host, so its companion is probably stellar. Two on-target duotransits survive. In TIC 159614322, a partial third transit leaves P=89.965 or 179.930 d; TIC 229055790 is the other. For 193 candidates that remain single, a median 52% of log-period space (5–1500 d) is now excluded.

Keywords: Exoplanet detection methods (489); Transit photometry (1709); Eclipsing binary stars (444)

Sample

ExoFOP lists hundreds of CTOIs uploaded as single transits, with no period, mostly in 2020–2022. We took the 450 open CTOIs flagged as single or duo transits, excluding those of groups still working their candidates. We then removed 46 with missing parameters, 8 on stars hosting confirmed planets, 4 on stars with a TOI, and 24 on catalogued TESS eclipsing binaries (Prsa 2022). This left 368 CTOIs on 359 stars. Of these, 32 already carry a community period and were not re-analyzed. Some entries are simply stale: CTOI 1042868.01, for example, is the RV-detected planet HD 88986 b, whose possible TESS transits were analyzed by Heidari 2024.

Methods

We used SPOC 2-min, TESS-SPOC (Caldwell 2020) and QLP (Huang 2020) light curves from MAST. For Sectors ≥100 without pipeline products, we performed our own aperture photometry on TESScut cutouts (Brasseur 2019); it recovers a known eclipse depth to 2%.

The repeat search slides a box of the catalogued depth δ and duration D on a D/4 grid, with segment-aware local baselines. A detection needs SNR ≥8, a depth of 0.5–2δ, at least 80% coverage, and baseline on both sides. Candidate periods come from linear-ephemeris fits to all events. Any period with a predicted epoch on flat data is rejected. We validated the method blind, starting from the first transit only, on 26 TESS duotransit hosts with known periods (including Hawthorn 2024). It kept the true period in 26/26 and returned it as the unique solution in 19, with no wrong unique answers.

We localized sources with TESScut difference images, measuring offsets from the target's Gaia DR3 position (Gaia 2023). A source is off-target if it is offset by ≥6” at ≥3σ; “firm” requires two agreeing images. On known planets the offsets are 1–3”. Binarity indicators are Gaia DR3 non_single_star, two-body orbits, and RV variability (p≤0.01, renormalised GOF >4; Katz 2023).

Results

Of the 306 stars, 34 do not show the catalogued event in our reductions, and 193 show no repeat. The 79 stars with repeats break down as follows:

• 21 are on Gaia-flagged binaries. Three Gaia SB1 orbits reproduce our TESS periods (Table 1), so these are eccentric eclipsing binaries.

• 18 are firmly off-target, i.e. blends with neighbors.

• 23 show repeats that no single period explains.

• 17 were vetted by hand: 2 survive (below), 3 remain tentative, 5 are possibly off-target, and 7 were rejected (inconsistent depths, absent in an independent reduction, or a host too large).

The 3 tentative cases have a repeat seen so far only in our own Sector 107 photometry.

CTOI 147035057.01. Three eclipses (Sectors 7, 33, and 108, the last from our own photometry) fix P=37.31497±0.00003 d; the S108 eclipse rules out every shorter alias. The eclipse is on target, with no secondary at phase 0.5. A fit with impact parameter and radius ratio free favours a V-shaped grazing eclipse over a full transit (ΔBIC ≈10). The radius ratio is k=0.34 (95%: 0.09–0.75), so R_2>1.7 R_J for the 2.0 R_{sun} host. Gaia DR3 measures an RV amplitude of 50 km s^{-1}, which implies M_2>~0.6 M_{sun}. The companion is probably stellar. A low RUWE (1.12), however, leaves a near-equal SB2 pair possible, and Gaia DR4 epoch RVs should decide.

TIC 159614322. Transits in Sectors 14 and 41 have consistent depths (≈3.2 ppt, ≈9.7 h) across products, and they are on target. Flat data leave P=719.72/n with n in {1,2,4,5,8}. Sector 21 then contains an ingress and ≈9 h of transit floor, with the egress lost in the orbit-1 downlink gap. It appears in TESS-SPOC, QLP, and our own photometry, with a flat background and no pointing step. A blind re-analysis with independent code also found it. It selects n=8 or n=4, giving P=89.9651±0.0006 or 179.9302±0.0012 d with T_0=BJD_{TDB} 2458703.8277±0.0041. These values are fitted from the two full transits; the partial transit arrives 13–26 min late, which may reflect a timing bias from the missing egress. The companion is ≈0.9 R_J, and the host is RV-stable in Gaia DR3. Both periods predict a transit at 2026-12-28 06:42 UTC (±45 min, errors inflated ×2.1). A dedicated sweep of all 272 stars for transits truncated by data gaps found no other convincing case: gap-edge dips do not outnumber gap-edge brightenings.

TIC 229055790. This candidate, uploaded by the Planet Hunters TESS project (see Eisner 2021), shows transits in Sectors 1 and 68 with consistent depths, on target, re-detected independently. With 38 allowed periods ≥26.7 d and R_p≈3.8 R_{Earth}, it is worth monitoring.

Table 1. Selected systems.

TIC	Events (Sectors)	Period	Notes
147035057	7, 33, 108(a)	37.31497±0.00003 d	grazing; Gaia RV-variable; likely EB
159614322	14, 41; 21 (partial)	89.9651±0.0006 or 179.9302±0.0012 d	≈0.9 R_J; on target; Gaia RV-stable
229055790	1, 68	1838.945/n≥26.7 d (38 aliases)	≈3.8 R_{Earth}; on target
231969683	6, 32, 34, 87, 98	21.978 d	Gaia SB1: 21.969±0.005 d
230130556	32 sectors, 15–86	22.259 d	Gaia SB1: 22.245±0.024 d
458419988	48, 75	734.56/40=18.36 d	Gaia SB1: 18.355 d
268534931	3, 30; 107(a,b)	92.60, 46.30, 30.87 or 23.15 d(b)	candidate of Hawthorn 2024
139726266	32; 107(a,b)	2109.0/n d(b)	≈2.8 R_{Earth}

(a) Our own TESScut photometry. (b) Tentative; to be rechecked when pipeline products for Sector 107 appear. The per-CTOI classification of all 306 stars is available in the repository cited below.

Conclusion

Stale single-transit CTOIs with repeat signals are mostly stellar, and inexpensive Gaia binarity checks and difference imaging dispose of most of them. The residue is small but useful: two on-target duotransits with alias lists short enough to target, and a few tentative repeats that pipeline products will confirm or reject. Code, light-curve products and classifications are at https://github.com/coracle-us/duo-transits.

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Facilities: TESS, Gaia

Software: lightkurve (Lightkurve 2018), astroquery (Ginsburg 2019), batman (Kreidberg 2015), astrocut (Brasseur 2019)

AI disclosure statement

The author used an AI assistant, Anthropic's Claude (a large language model), under his supervision. It wrote and ran the analysis code and drafted and edited the text of this note. The author reviewed and edited the text, checked the numbers against the saved analysis outputs, and takes full responsibility for the content. Every reference was checked against ADS and arXiv records. The note contains no AI-generated figures.

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